INDUSTRIAL INTERNSHIP PROJECT BRIEF

Project Type: Industrial Internship leading to Cash Prize Incentives
Industry Partner: Burk.pk (Powered by Devomech Solutions Pvt. Ltd.)
Project Title: Automated Supply Chain & Smart Order Verification System (SCM-OVS)

---

1. Project Overview

Burk.pk is an active e-commerce brand seeking to digitize, fully automate, and error-proof its internal supply chain and fulfillment framework. Currently, managing multi-channel customer orders (coming from both our independent platform **Burk.pk** and our marketplace storefront on **Daraz.pk**) introduces significant manual bottlenecks, human errors in packing, and communication delays.
This project aims to develop an intelligent, central software utility that aggregates orders across multiple sales channels, provides real-time inventory and supplier management, integrates physical hardware verification (barcode scanners/cameras) at the warehouse packaging stations to ensure zero-error fulfillment, and automates tracking and follow-up pipelines. Successful completion of this project will lead directly to industrial internship opportunities at Burk.pk and an exclusive cash prize for high-performing teams.

2. Project Incentives & KPIs

To foster academic excellence and reward professional execution, Burk.pk offers the following performance-based rewards to the selected project member:

- **Performance Cash Prize:** If the student meets all specified system requirements and achieves the established Key Performance Indicators (KPIs) upon final delivery, a cash prize up to **PKR 100,000 (One Hundred Thousand Rupees)** will be awarded to the team.

Key Performance Indicators (KPIs) for the Reward:

1. **Zero-Mismatch Validation:** The barcode validation mechanism must effectively intercept and visually/audibly flag 100% of mismatched inventory items scanned during test packaging loops.
2. **Synchronous Communication Latency:** Automated media generation (packaging recording) and dispatch via WhatsApp/Email must initiate successfully within 3 minutes of an order being sealed and closed out by the operator.
3. **Multi-Channel Sync Accuracy:** API loops fetching multi-platform data from both Burk.pk and Daraz.pk must reconcile order metrics without record dropped or duplicated entries.

3. Core Project Modules & Functional Requirements

Module 1: Centralized Inventory & Supplier Management

- **Real-time Stock Tracking:** Monitor stock levels uniformly across all storage categories.
- **Smart Reorder Alerts:** Trigger proactive system alerts when any component falls below its designated safety-stock threshold (Low Inventory Warning).
- **Supplier Directory integration:** Store full supplier profiles including vendor contact parameters, wholesale pricing tiers, and delivery lead times. Provide a mechanism to immediately place or draft a new supply order directly from the dashboard when inventory runs low.

Module 2: Multi-Channel Order Aggregation

- **Unified Order Pipeline:** Connect to external storefront endpoints to automatically detect and pull incoming orders from **Daraz.pk** (via Daraz Seller Center API integration or web data ingestion loops) and **Burk.pk** into one combined, live queue.

Module 3: Intelligent Fulfillment & Barcode Verification

- **Digital Packing List Generation:** Build a system interface that outputs clear packing checklists prioritized by receipt time or courier deadlines.
- **Anti-Error Scanner Integration:** Connect physical barcode/QR scanners directly to the software logic. Operators must scan each item prior to sealing it in the package.
- **Real-Time Error Highlighting:** If an operator scans an item that does not belong to that active order, the interface must instantly flash a highly visible error warning and emit an audio signal to block the incorrect item from being packed.

Module 4: Automated Video Surveillance & Customer Transparency

- **Packaging Smart Camera Capture:** Trigger a localized overhead workspace camera to record the full physical packaging sequence of each specific order.
- **Automated Media Dispatch:** Upon finalizing and closing out the order packaging routine, the platform must process/compress the clip and forward it directly to the customer via automated **Email** and **WhatsApp** channels.

Module 5: Logistics Integration & Automated Tracking

- **Courier API Linkage:** Integrate with regional shipping and logistics services (e.g., Leopard, Trax, TCS, or M&P).
- **Instant Airway Bill Booking:** Automatically push shipping information to the logistics partner's server to generate courier bookings instantly without manual text data entry.
- **Tracking Code Distribution:** Retrieve the resulting tracking ID back from the courier api and broadcast it directly to the end customer via **Email** and **WhatsApp**.

Module 6: Post-Purchase Engagement Loop

- **Delivery Review Trigger:** Monitor the courier platform webhooks or delivery status. Once the item updates to "Delivered," the system will initiate a pre-timed automated notification across WhatsApp and Email reminding the user to submit their platform product review and feedback.

4. Recommended Technical Stack

While the academic board or students can propose equivalent technology alternatives, the recommended structure includes:

- **Backend Services:** Python (Django or FastAPI) or Node.js (Express), selected for structural API manipulation capabilities.
- **Frontend Interface:** React.js, Next.js, or Vue.js to develop a fast, accessible inventory dashboard for warehouse personnel.
- **Computer Vision & Video Pipelines:** OpenCV frameworks or FFmpeg hooks to manage streaming data, clipping, and media file optimization.
- **Databases:** PostgreSQL or MongoDB to archive system state, customer listings, logs, and product directories.
- **Messaging APIs:** WhatsApp Business Cloud API / Twilio for mobile notifications and SendGrid/Nodemailer for robust mailing operations.